

locking hinge to provide the collapsibility required without compromising stability. The design also integrated a suction cup/pump mechanism to allow for fast, repeatable attachment to the x-ray panels. Sounds complicated, right?

Three days later, the SGW team delivered the first physical prototypes to the client. The prototypes were machined out of aluminum directly from the 3D model, using a technique called computer numerical control (CNC) and were hand assembled by the SGW team.

The client immediately took the prototypes to its military contact for review. The general concept was accepted with a number of minor design modifications. In the next five days, another full cycle of design iteration, prototyping, and design review was completed by the client and SGW. The first production run of forty completed units was ready for delivery three and a half weeks after the initiation of the development project.

SGW realized that this was a winning model because feedback on design decisions was nearly instantaneous. The team used the same process to design and deliver eight products, serving a wide range of functions, in a twelve-month period. Half of those products are generating revenue today, and the rest are awaiting initial orders, all thanks to the power of working in small batches.

THE PROJECT TIME LINE	
Design and engineering of the initial virtual prototype	1 day
Production and assembly of initial hard prototypes	3 days
Design iteration: two additional cycles	5 days
Initial production run and assembly of initial forty units	15 days

Small Batches in Education

Not every type of product—as it exists today—allows for design